



Assignment No: 02

Marks: 80

Section: BCS (A to J)

CLO: 01

Deadline: 02-03-2026 before 02:00pm

Semester: Spring-26

Instructions:

1. Plagiarized work will result in zero marks.
 2. No retake or late submission will be accepted.
 3. Attach complete code, results, and screenshot for questions that require programming solutions. Programs/codes should be typed.
 4. **Complete assignment is to be submitted in soft-copy as well as in hard-copy. Submit the hardcopy before the deadline through CR, and softcopy on GCR.**
 5. The softcopy should be a single PDF file of your complete assignment including programming and non-programming questions.
 6. The PDF file should be according to the following format: id_section_A1 e.g., i25-123456_A_A2. A2 in the end denotes Assignment 2. **The title page must include complete student information, including name, section, id, course name, and assignment number.**
 7. The images of a by-hand solution should be properly scanned. You can use any mobile application such as CamScanner or Adobe Scan for scanning. Each of these applications allows you to export pdf or image files which you can use to combine with your programming solutions. Do not attach direct images from the camera application of your mobile phone, or screenshots.
 8. Python is the only approved programming language.
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Part 1: Logistics & Navigation Systems (Norms & Metrics, 12.5)

Problem 1: The "Spider-Bot" Logistics Unit

You are programming the navigation system for a "Spider-Bot" maintenance unit on the hull of a space station. The hull surface is a grid of solar panels. The bot is currently at $A(1, 2, 3)$ and must reach a damaged panel at $B(5, 8, 9)$.

The bot has three distinct movement modes, each governed by a different mathematical norm:

1. **Hover Mode (L_2 Norm):** Uses thrusters to fly in a direct straight line through space (fuel expensive).
2. **Crawl Mode (L_1 Norm):** Moves strictly along the grid lines of the panels (cannot move diagonally).
3. **Crane Mode (L_∞ Norm):** Extends hydraulic arms simultaneously in x, y, z directions. The time taken is determined by the *longest* single extension required.

Your Tasks:

1. **Cost Analysis:** Calculate the "cost" (distance) for all three modes to travel from A to B:
 - d_2 (Euclidean distance)
 - d_1 (Manhattan/Taxicab distance)
 - d_∞ (Chebyshev distance)
2. **Critical Inference:** If the bot has only 7 units of fuel (where 1 unit distance = 1 unit fuel), which mode(s) are impossible to use?
3. **Python Task:** Write a Python script that accepts any two points P_1 and P_2 as input and outputs the cost for all three modes, identifying the most efficient one.

Problem 2: The Laser Defense Grid

A debris fragment is detected moving through the sector. Its trajectory is a straight line L_1 passing through points $P(1,0,1)$ and $Q(3,3,2)$. To protect the station, you activate a laser defense wall (a plane). The plane is determined by a laser emitter at the origin $O(0,0,0)$ and two direction vectors $u = \langle 1, -1, 0 \rangle$ and $v = \langle 0, 1, 1 \rangle$. To reinforce the defense, a secondary "Backup Shield" is generated parallel to the first. This new shield is defined by the equation $x + y - z = 10$.

Your Tasks:

1. **Trajectory Mapping:** Find the parametric equation of the debris path (L_1) and the scalar equation of the Laser Plane.
2. **Collision Course:** Will the debris hit the Laser Plane? Find the exact coordinates of the intersection.
3. **Sensor Safety:** A critical sensor is located at $S(5, 5, 5)$. Calculate the shortest distance from the sensor to the debris path (L_1) to ensure the sensor is not damaged by the wake.

4. **Multi-Object Tracking:** A second debris fragment is moving along line L_2 with equation $x = 2t$, $y = 4t$, $z = 0$. Determine if the two debris paths (L_1 and L_2) will collide (intersect) or if they are skew lines. If they are skewed, calculate the minimum distance between them. (*Skew lines are two non-parallel lines in three-dimensional space that do not intersect and are not coplanar*)
5. **Shield Clearance:** The two shields are parallel. Calculate the perpendicular distance between the main Laser Wall (from Task 1) and this Backup Shield to ensure a fighter jet can fly safely between them.
6. **Safe Zone:** A repair drone is hovering at $D(2, 3, 0)$. Calculate the shortest distance from this drone to the Backup Shield to avoid accidental incineration.

Part 2: Surface Analysis & Precision Engineering (14.6)

Problem 3: The Solar Concentrator Array

A concentrated solar power plant is being built at a mountainous region whose terrain is modeled by the surface $x^2 + 2y^2 + z - 10 = 0$. There is a massive solar receiver located at $P_0(1, 2, 1)$.

Your Tasks:

1. **Drone Docking Vector:** A maintenance drone needs to dock with the receiver. To maximize efficiency, it must approach along the normal line to the terrain at that exact point. Derive the parametric equations for this path.
2. **Structural Foundation:** A support beam must be bolted to the mountain at P_0 . This beam must lie perfectly flat against the tangent plane to ensure structural stability. Determine the equation of this plane.
3. **The Intersection:** A secondary sensor cable follows the path $\mathbf{r}(t) = \langle t, 2t^2, 11 - 10t \rangle$. Determine if this cable is tangent, normal, or neither to the mountain's surface, when it passes through the point where $t = 1$. Justify your answer using the gradient vector ∇f .

Problem 4: The Bio-Reactor Sensitivity Analysis

A pharmaceutical company is designing a cylindrical mixing vat for vaccine production. The target dimensions are radius $r = 2\text{m}$ and height $h = 10\text{m}$. The volume is $V(r, h) = \pi r^2 h$. Due to manufacturing tolerances, there is an uncertainty of $\pm 0.01\text{m}$ in the radius and $\pm 0.05\text{m}$ in the height.

Your Analysis:

1. **Sensitivity Analysis:** Analytically determine whether the vat's volume is more sensitive to a slight change in the radius or a minor change in the height at the target dimensions.
2. **Error Estimation:** Use the total differential ∂V to estimate the maximum possible error in the calculated volume resulting from these uncertainties.

Problem 5: The "Perfectly Flat" Mirage

A precision sensor is placed at $P_0(0, 0, 1)$ on a cooling plate modeled by $f(x, y) = \cos(x) \cdot \cos(y)$. You need to predict the temperature at a nearby point $(0.1, 0.1)$ using the linearization $L(x, y)$.

The Theoretical Challenge: Without calculating the error bound M (the second derivatives), explain why the linearization $L(x, y)$ will always overestimate the true value of $f(x, y)$ for any point in a small neighborhood around $(0,0)$.

Problem 6: Terrain Mapping & Drone Safety

You are developing software for a drone landing on uneven terrain defined by $f(x, y) = \sin(x) \cdot \cos(y) + 0.1x^2$. The drone must land at $P_0(2, 1)$.

Python Task: Write a Python script to:

1. **Define the Surface:** Create a 3D plot of the terrain over $x \in [-2, 4]$ and $y \in [-2, 4]$.
2. **Numerical Gradient:** Calculate the partial derivatives f_x & f_y at P_0 using finite difference method (small h).
3. **Visuals:** Plot the tangent plane at P_0 using gradient and draw the normal vector to it starting at P_0 .

Part 3: Optimization & Stability Analysis (14.7)

Problem 7: CPU Thermal Stress Test (Constraint-Bounded Optimization)

You are analyzing the heat distribution on a square metallic microprocessor plate. The energy density is given by $E(x, y) = x^3 - 3x + y^3 - 3y + xy$ over the region $R = \{(x, y) | -2 \leq x \leq 2, -2 \leq y \leq 2\}$.

Your Tasks:

1. **Internal Hotspots:** Compute E_x and E_y and solve $E_x = 0$ and $E_y = 0$. Identify all interior critical points where the heat gradient is zero .
2. **Stress Analysis (Hessian Matrix):** Use the Second Derivative Test to classify these points. Identify any local minima, local maxima, and/or saddle points, as these represent structural stress risks where the material expands in one direction and contracts in other. Do notify in case the test fails at any pint.
3. **Saddle Point Verification:** Verify the behavior of function in case of saddle point (if any) with reasoning (e.g., function increases in one direction and decreases in other)
4. **Boundary Failure Analysis:** The edges of the chip are connected to the frame. After analyzing interior critical points, analyze the heat by evaluating E along the four edges ($x = \pm 2, y = \pm 2$) to find the Absolute Maximum and Absolute Minimum Heat values & location on the entire component.

5. **Boundary Edge Test:** Select one edge and apply first derivative test in one variable, classify local extrema along that boundary and explain whether it remains local relative to the full region R.
6. **Python Computational Verification:** Use SymPy to define $E(x, y)$, compute first and second partial derivatives, solve for critical points symbolically, compute the Hessian determinant at each critical point, evaluate E at interior and boundary candidates, and determine computationally the absolute maximum and minimum.

Problem 8: Chemical Reaction Stability (Multi-Variable Optimization)

You are controlling a chemical reaction where the system's energy state E depends on three control variables. The energy function is:

$$z = -x^3 + 3xz + 2y - y^2 - 3z^2$$

Note: A local minimum represents a stable state, while a maximum or saddle point represents instability.

Your Tasks:

1. **Equilibrium Detection:** Find the critical point(s) (x, y, z) where the first partial derivatives are all zero.
2. **Stability Test (Hessian Matrix):** Construct the 3×3 Hessian Matrix $|H|$. Calculate the three principal minors to determine if the matrix is positive definite, negative definite, or indefinite.
3. **Verdict:** Is the reaction at this critical point stable (minimum), unstable (maximum), or metastable (saddle)?